

# Progress Report: Repository Update, Evaluation-Criterion Validation, and Weight-PCA (Predictor 6)

*Thesis: A Unified Benchmark of Grokking Predictors in Neural Networks*

Student: Jonathan John | Supervisor: Sheikh Faisal Rashid | Period covered: 26 September – 2 October 2026

## Overview

The repository, including all results and model checkpoints, was pushed to GitHub. Two errors in the Spectral predictor's (Predictor 3) explanation were found and corrected, with no change to its verdict. The evaluation criterion used to score every predictor was independently validated against synthetic signals with known correct answers. The Correlation Traps predictor was removed from the evaluation order, since it targets a different phenomenon (post-grok collapse) than this benchmark tests for (pre-grok onset). Predictor 6, Weight-PCA, was implemented, run, and closed negative. Six of the eight predictors now in the evaluation order are closed, all negative.

## 1. Objectives completed

- Push the repository, including all results and model checkpoints, to GitHub.
- Correct two errors found in the Spectral predictor's (Predictor 3) explanation.
- Validate the evaluation criterion itself using synthetic controls, independent of any real predictor.
- Resolve the Correlation Traps scope question and update the evaluation order.
- Implement and score Predictor 6, Weight-PCA.

## 2. Work accomplished

**(a) Repository pushed to GitHub.** The repository, including the full results folder and all 4005 model checkpoints (about 3.2 GB), is now on GitHub. One oversized commit (about 2.8 GB) was split into five smaller commits, one per seed, so the upload would complete reliably; no history was rewritten and nothing was deleted.

**(b) Spectral predictor (Predictor 3) - explanation corrected.** On review, two statements in the earlier write-up were wrong. First, the  $k_{90}$  statistic (the number of spectral modes needed to capture 90% of target power) is not meaningful here, because the representation kernel's rank cannot exceed 128 (the hidden-layer width):

$$\text{rank}(K) \leq 128 \text{ (hidden width)}, \quad \text{PR} = \frac{\left( \sum_{k=1}^{50} \eta_k \right)!^2}{\sum_{k=1}^{50} \eta_k^2}$$

so  $k_{90}$  always falls inside a numerically arbitrary null space. Second, effective dimension is correctly measured by the participation ratio PR above, taken over the top 50 eigenvalues, and this value does fall sharply close to the grok epoch - so the earlier claim of "no rank collapse" was incorrect. **The predictor's verdict is unchanged (CLOSED, NEGATIVE); only the explanation is corrected.**

**(c) Evaluation criterion validated with synthetic controls.** Before trusting any further predictor verdicts, the pass/fail rule itself was tested against five synthetic signals with known correct answers (an ideal predictor, a constant signal, a post-grok signal, a noisy-good predictor, and two random-predictor baselines):

$$\text{lead\_frac} = \frac{1}{n} \sum_i \mathbb{1}[P_i < G_i - m], \quad \text{mean\_rel\_lead} = \frac{1}{n} \sum_i \frac{G_i - P_i}{G_i}$$

$$\text{PASS if: } \text{lead\_frac} \geq 0.8 \quad \wedge \quad \rho \geq \rho_{\text{thresh}} \quad \wedge \quad 0.05 \leq \text{mean\_rel\_lead} \leq 0.9$$

The correlation threshold  $\rho_{\text{thresh}}$  was calibrated from data (the 99th percentile of 10,000 random predictors' rank correlation with the real grok epochs), giving  $\rho_{\text{thresh}} = 0.90$ . Full results are in Section 4.

**(d) Correlation Traps removed from the evaluation order.** Re-checking the source paper (Prakash and Martin) showed that Correlation Traps is defined to detect a collapse back to chance performance well after a model has already grokked - the opposite of this benchmark's question, which is whether a signal leads the onset of grokking. It has been removed from the evaluation order rather than tested against the wrong target. The order is now: L2 Norm,

Dropout, Spectral, AGE, HTSR Alpha, Weight-PCA, Higher-MI, Commutator Defect (eight predictors, down from nine).

**(e) Predictor 6, Weight-PCA, implemented and run.** Weight-PCA (Yunis et al., 2024) measures the effective rank of each weight matrix  $W$  via the entropy of its normalised singular-value distribution ( $R$  is the matrix's maximum possible rank):

$$\text{NormEffRank}(W) = \frac{-\sum_i p_i \log p_i}{R}, \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j}$$

It was run checkpoint-only against the existing 5-seed run (401 of 801 saved checkpoints per seed); no retraining was needed. Full results are in Section 4.

### 3. Challenges encountered

- Pushing about 3.2 GB of checkpoints to GitHub required splitting one oversized commit by seed; resolved without any loss of data or history.
- The Spectral predictor's  $k_{90}$ -based explanation (Section 2b) was found to be wrong on review and had to be corrected; caught before it was used anywhere else.
- One of the five synthetic controls used to validate the evaluation criterion (a random predictor that always fires before grok, by construction) has a false-positive rate of 21.7%, above the 5% target. The cause is structural - that control's random range scales with the grok epoch itself - not a flaw in the criterion as applied to real predictors (the real-world random-predictor case has a false-positive rate of only 0.22%). A proper fix would need more seeds (about 7 hours of extra training); not done this period, and recorded as an open limitation.
- A misreading of the Correlation Traps paper (Section 2d) was caught before any predictor code was written for it, avoiding wasted implementation effort.

### 4. Results obtained

**Evaluation-criterion validation (synthetic controls,  $\rho_{\text{thresh}} = 0.90$ ).**

| Control                                     | Meant to              | Pass rate | Verdict                   |
|---------------------------------------------|-----------------------|-----------|---------------------------|
| A - Ideal predictor                         | Pass 100%             | 100%      | OK                        |
| B - Constant signal                         | Fail 0%               | 0%        | OK                        |
| C - Post-grok signal                        | Fail 0%               | 0%        | OK                        |
| D - Noisy-good (25% noise)                  | Degrade gracefully    | 91.0%     | OK                        |
| D - Noisy-good (100% noise)                 | Degrade gracefully    | 21.9%     | OK                        |
| E (v1) - Random, unconditional              | $\text{FPR} \leq 1\%$ | 0.22%     | OK                        |
| E (v2) - Random, always leads (stress test) | $\text{FPR} \leq 5\%$ | 21.7%     | Above target - structural |

**Predictor 6, Weight-PCA - grok epochs: 12987 / 6921 / 9510 / 11019 / 23747.**

| Seed | Grok epoch | NormEffRank: first → last | Min epoch (× grok) | Min value |
|------|------------|---------------------------|--------------------|-----------|
| 0    | 12987      | 0.0373 → 0.0166           | 37100 (2.86×)      | 0.01636   |
| 1    | 6921       | 0.0373 → 0.0186           | 27900 (4.03×)      | 0.01823   |
| 2    | 9510       | 0.0373 → 0.0185           | 32400 (3.41×)      | 0.01750   |
| 3    | 11019      | 0.0373 → 0.0174           | 29700 (2.70×)      | 0.01695   |
| 4    | 23747      | 0.0373 → 0.0182           | 36200 (1.52×)      | 0.01755   |

Frozen-protocol scoring (same two tests as Predictors 1-5):  $\text{lead\_frac} = 0.000$  (need  $\geq 0.8$ );  $\rho = 0.800$  (need  $\geq 0.900$ );  $\text{mean\_rel\_lead} = -1.903$  (need between 0.05 and 0.9). **Verdict: FAIL - CLOSED, NEGATIVE.** NormEffRank declines steadily through training in every seed, but its minimum is reached well after grok (1.52 to 4.03 times the grok epoch), confirming the source paper's own framing that rank collapse coincides with grokking rather than leading it.

**Consolidated predictor status (6 of 8 complete; Correlation Traps removed).**

| # | Predictor         | Status                                                       |
|---|-------------------|--------------------------------------------------------------|
| 1 | L2 Norm           | CLOSED - negative                                            |
| 2 | Dropout           | CLOSED - negative                                            |
| 3 | Spectral          | CLOSED - negative (explanation corrected; verdict unchanged) |
| 4 | AGE               | CLOSED - negative                                            |
| 5 | HTSR Alpha        | CLOSED - negative                                            |
| 6 | Weight-PCA        | CLOSED - negative (this period)                              |
| 7 | Higher-MI         | Next - not started                                           |
| 8 | Commutator Defect | Not started                                                  |

Six predictors in a row now close negative under one fixed, pre-registered protocol. This does not mean the original papers are wrong in their own settings; it means their signals have not, so far, transferred as ex-ante early-warning indicators for grokking under one unified, stricter evaluation.

## 5. Plans for the next reporting period

- Implement and run Predictor 7 (Higher-MI), then Predictor 8 (Commutator Defect), completing the eight-predictor benchmark.
- Persist the frozen-protocol verdict (lead\_frac, p, mean\_rel\_lead, pass/fail) to a results file for each predictor; it is currently printed to the console only.
- Recompute the Spectral predictor's corrected figures (the k\_90 null-space percentages and participation-ratio values) with a dedicated script before they are used in the thesis.

**For the next supervision meeting:** the decision to remove Correlation Traps from the evaluation order; the evaluation criterion's documented false-positive limitation (Section 3); and whether six (soon eight) consecutive negative verdicts should already begin shaping the results chapter.